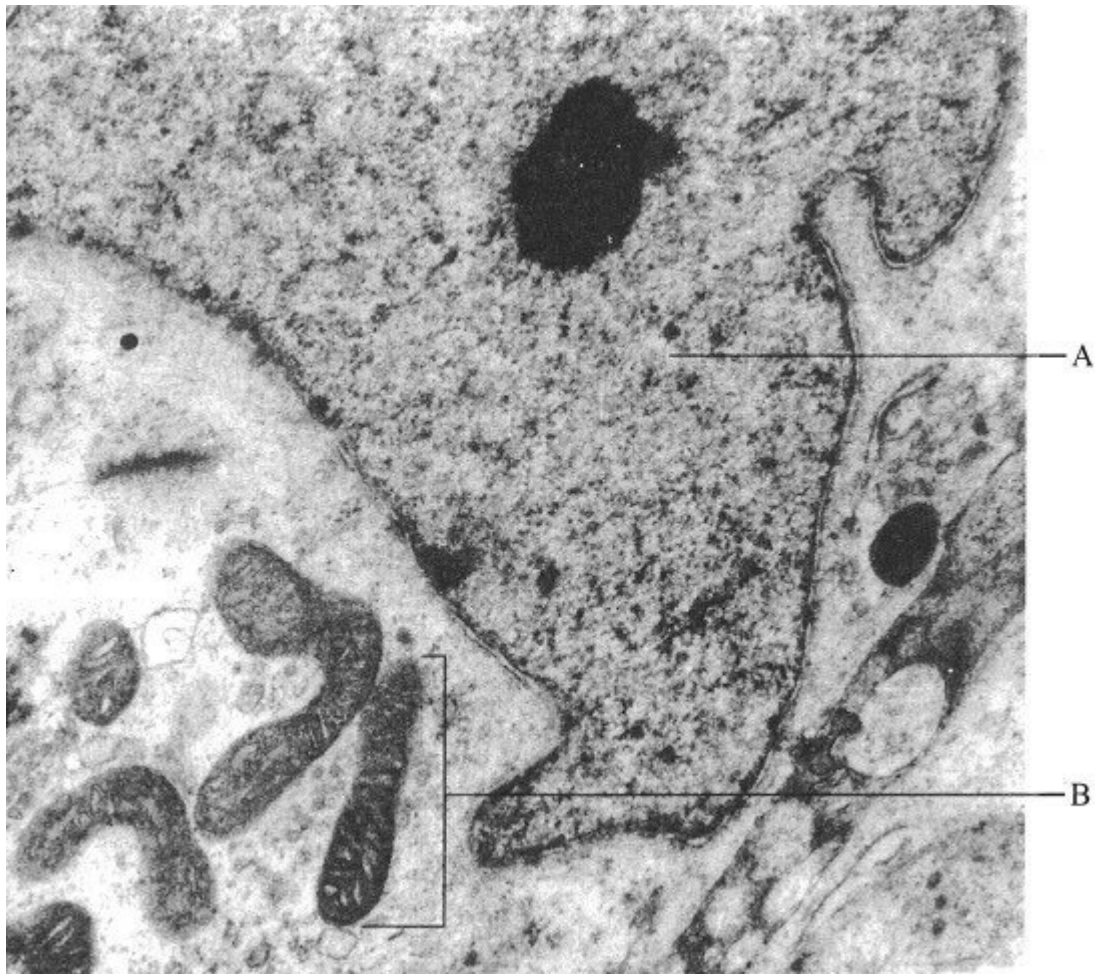


1. The photograph below-shows part of an animal cell, as seen using an electron microscope. The magnification is $\times 5000$.



What are the functions of A and B and what is the actual length of B?

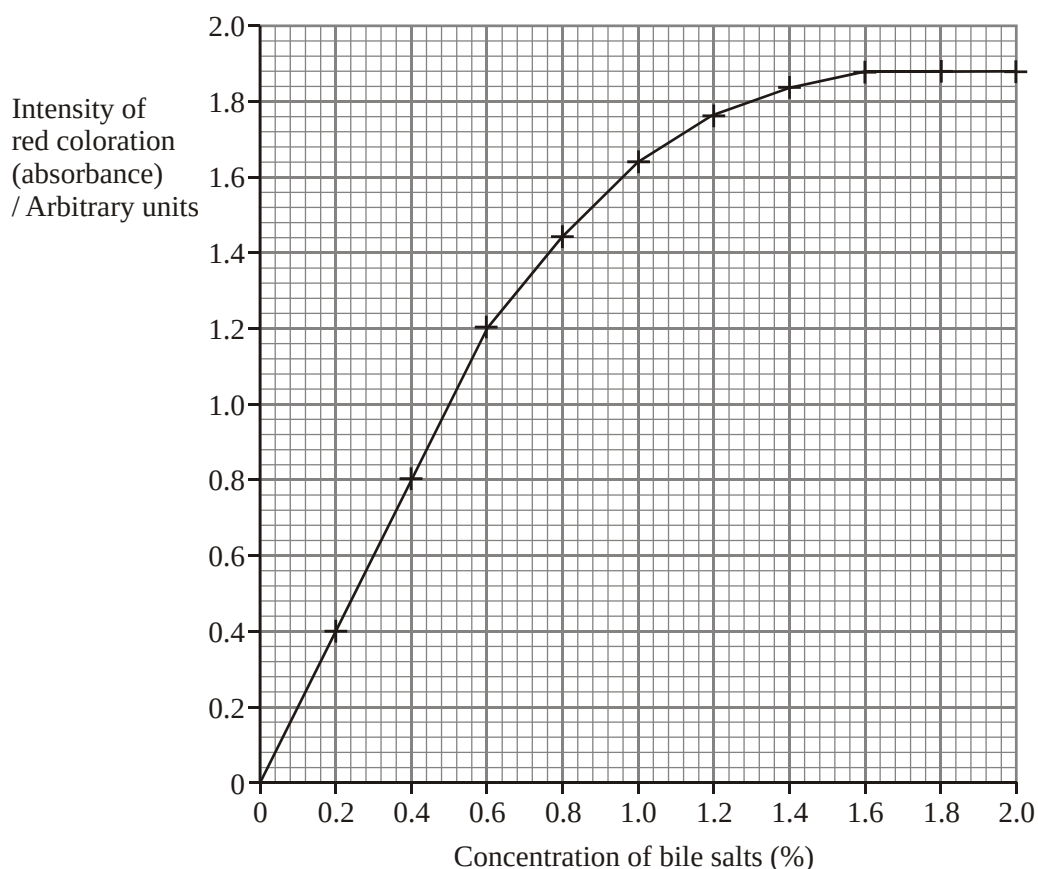
	A	B	Actual length of B
A	Regulates cell activity	Final chemical modification, sorting and packaging of proteins	72.0 μm
B	Maintains internal hydrostatic pressure within the cell	Post-translational modification	7.2 μm
C	Regulates cell activity	Site of ATP synthesis	7.2 μm
D	Maintains internal hydrostatic pressure within the cell	Site of lipid synthesis	72.0 μm

2. Beetroots are root vegetables. They appear red because their cells contain a water soluble red pigment in their vacuoles, which cannot pass through membranes.

Five beetroot discs were then placed in a tube containing 25 cm³ of 2% bile salt solution and left for 30 minutes at 20 °C. The procedure was repeated for different concentrations of bile salts and one set of discs was left in distilled water.

After 30 minutes, each set of beetroot discs was removed from the solutions and from the water. Each bile salt solution was stirred and a sample removed and placed in a colorimeter. The intensity of red coloration (absorbance) was determined by the colorimeter.

The results of the investigation are shown in the graph below.

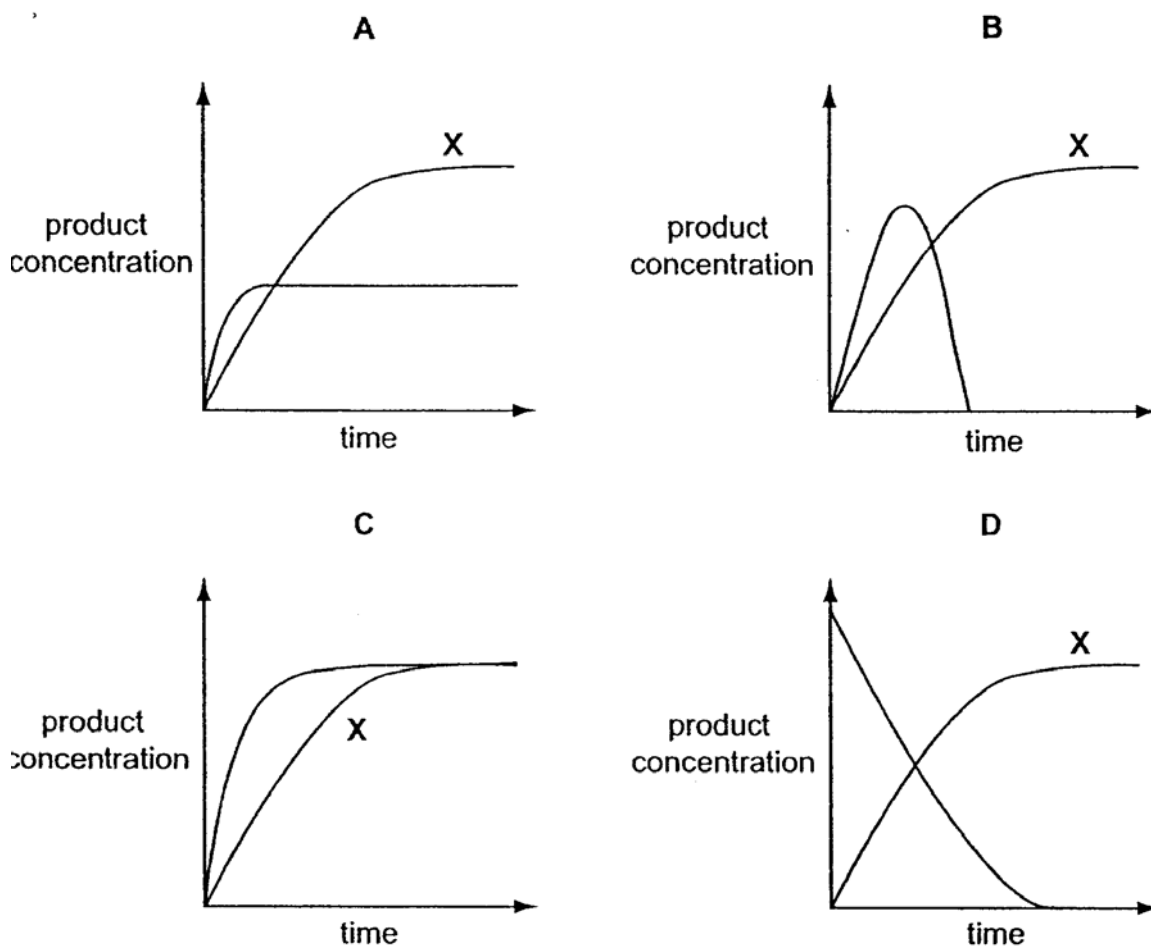


What is the best explanation for the results?

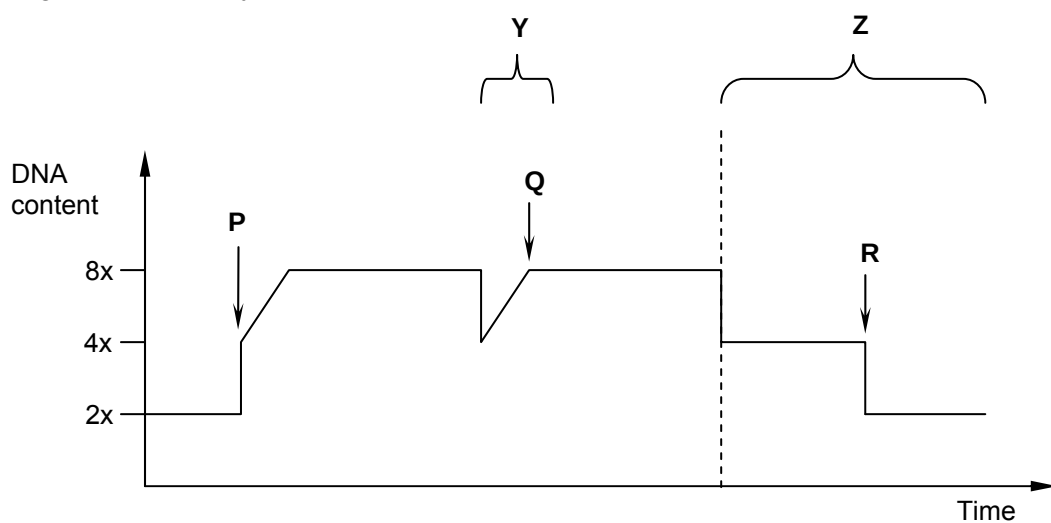
- A Pigments leaked out via protein channels in the plasma membrane.
- B Bile salts caused phospholipids in both tonoplast and plasma membrane to be emulsified.
- C Bile salts did not have any effect on the tonoplast.
- D Temperature affected the intensity of colour absorbance.

3. The enzyme amylase hydrolyses starch into disaccharides. Amylase is unable to breakdown cellulose because
- A cellulose does not contain glycosidic bonds.
 - B starch consists of glucose; cellulose consists of other monosaccharides.
 - C the bonds between sugars in cellulose are much stronger.
 - D the sugars in cellulose are bonded together differently than in starch, hence cellulose have a different shape.
4. Two enzyme experiments were carried out. The first, experiment X, was carried out at a constant temperature of 37°C. During the second experiment the temperature was increased from 37°C to 80°C.

Which graph shows the results?



5. The graph represents the changes in the DNA content within a cell at different stages in the cell cycle.



Name the events occurring at **P**, **Q** and **R**, and identify the stage where meiosis was occurring.

	P	Q	R	Meiosis occurs at
A	S phase	Fertilisation	Cytokinesis	Y
B	Fertilisation	Interphase	Cytokinesis	Z
C	S phase	Prophase	Telophase	Y
D	Fertilisation	Metaphase	Cytokinesis	Z

6. Which of the following processes will **NOT** be carried out to completion in a triploid cell?

- I** Metaphase I
- II** Prophase I
- III** Interphase
- IV** Anaphase

- A** I and II only
- B** III and IV only
- C** II and IV only
- D** I, II and IV only

7. The table below shows some amino acids and their DNA genetic code.

Amino acid	DNA code
Alanine	CGG
Lysine	TTT
Arginine	GCG
Phenylalanine	AAA
Glycine	CCA
Valine	CAA

A DNA sequence codes for a polypeptide arginine-glycine-phenylalanine-valine-alanine.

The following mRNA molecule was transcribed from the DNA sequence above.
Which mRNA triplet contains a single transcription error?

CGCGUUUUUGUUGCC

- A** the first
- B** the second
- C** the third
- D** the fourth
8. Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein product of that gene?

	Promoter	Terminator	Start codon	Stop codon	Middle of an intron
A	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced

9. Five different amino acids (numbered **1** to **5** below) form the following sequence in part of a polypeptide chain:

2-----3-----4-----1-----3-----5-----2

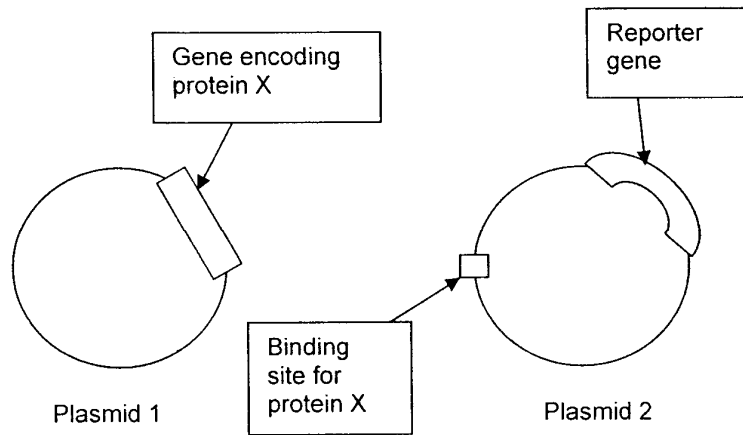
Messenger RNA (mRNA) codons which correspond to these amino acids are:

amino acid 1	UGU
amino acid 2	GAU
amino acid 3	CAC
amino acid 4	CAA
amino acid 5	AAG

Which one of the following DNA base sequences could provide the code for the given section of polypeptide?

- A** CTAGTGGTTACAGTGTTCTTA
 - B** CTAGTGGTTACAGTGTTGCTA
 - C** CTAGTGGTTTCTGTGTTGCTA
 - D** CTAGTGGTTACAGTGTTCTT
10. Genetic drugs are short sequences of nucleotides. One example is an anti-sense drug made from RNA nucleotides that are complementary to the protein-coding mRNA.
- Which is the process that is affected by the anti-sense drug and what is its most probable mode of action?
- A** It prevents DNA replication by binding to the anti-sense strand of DNA that codes for the disease-causing protein.
 - B** It prevents transcription by binding to the sense strand of DNA that codes for the disease-causing protein.
 - C** It prevents translation by binding to the mRNA that contains information for the disease-causing protein.
 - D** It prevents the normal functioning of disease-related protein when it is translated into a protein that inhibits the disease-causing protein.

11. An investigator wanted to determine the function of protein X. He cloned the gene X into plasmid 1. He then set up another plasmid (plasmid 2) with a binding site for the product of gene X plus a reporter gene. A reporter gene codes for the synthesis of a product that is easily measured. The two plasmids (shown below) were then introduced into the nucleus of a host cell.

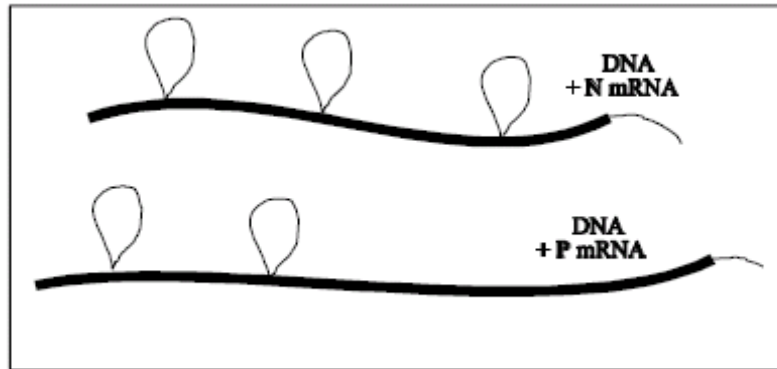


He noted that there was an increase in the number of reporter gene transcripts. Which of the following is a valid conclusion regarding gene X or product X?

- A Gene X is an enhancer sequence.
- B Gene X is a promoter sequence.
- C Product X is DNA polymerase.
- D Product X is a transcription factor.

12. People with the recessive disease, Purple Tongue Syndrome, are unable to make the enzyme Blue Pigment Decolorase (BPD). You isolate mRNA from a person (**P**) afflicted with the disease and from a normal person (**N**), allow them to form DNA-RNA hybrid heteroduplexes with the gene for BPD.

The diagram represents what is seen by electron microscopy. Double-stranded regions are represented by thick lines, and single-stranded regions are represented by thin lines.



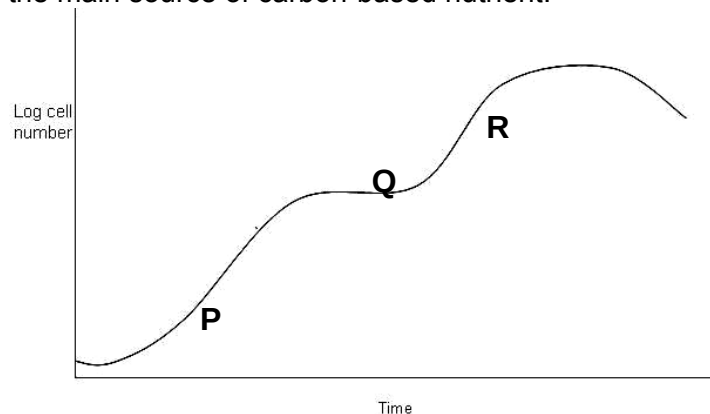
Which of the following could be a possible molecular cause of Purple Tongue Syndrome?

- A Failure of spliceosome to recognize splice sites leading to excision of the wrong intron
 - B Point mutation in the branch site leading to failure to excise one intron
 - C Gene amplification of a repressor protein involved in BPD gene regulation
 - D Expansion of a trinucleotide repeat in an intron causing splice site to be misread as the reading frame is shifted
13. Hepatitis C virus (HCV) can only replicate by injecting its genome into human cells upon infection. HCV is a positive-sense RNA virus i.e. genomic materials are RNA which can be used for translation immediately. Which of the following is most likely missing in the viruses like HCV?

- 1. DNA-dependent RNA polymerase
- 2. Reverse transcriptase
- 3. Ribosomes
- 4. RNA-dependent DNA polymerase
- 5. RNA-dependent RNA polymerase

- A 1 & 2
- B 1 & 3
- C 2 & 5
- D 3 & 5

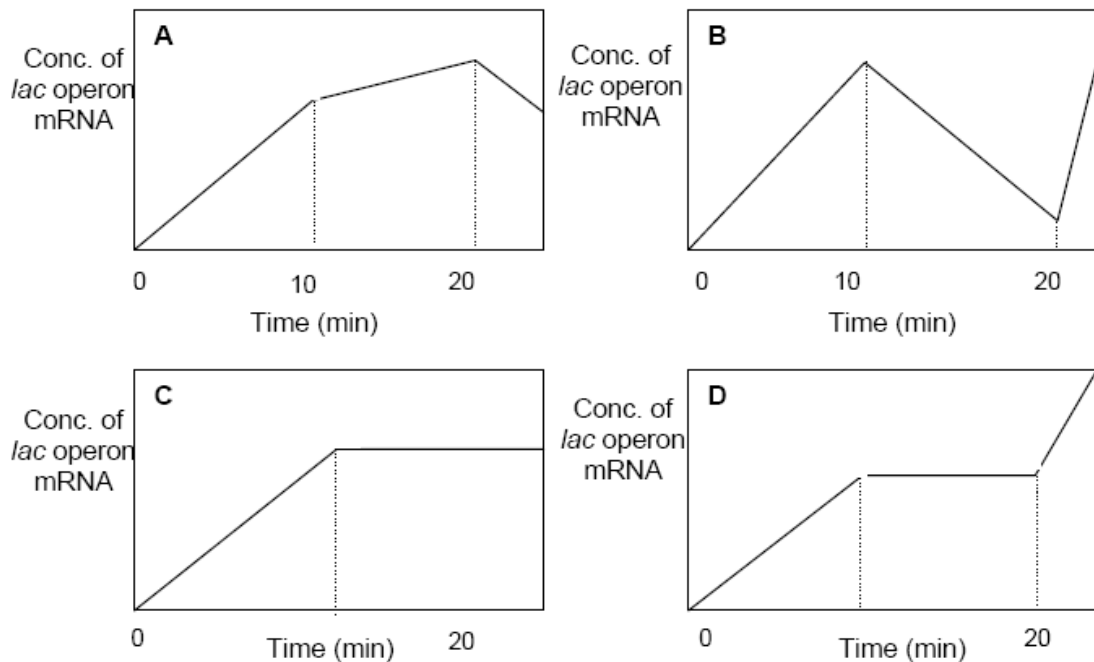
14. Oseltamivir in Tamiflu prevents detachment of influenza virus from infected host cell. Which protein function does oseltamivir inhibit?
- A gp 120
- B hemagglutinin
- C gp 40
- D neuraminidase
15. If 2 populations of genetically different bacteria are cultured in a U-shaped tube separated by a membrane filter with a pore size 8nm in length, but recombination takes place anyway, the mechanism of genetic exchange is by
- A generalised transduction
- B specialised transduction
- C conjugation
- D transformation
16. *E.coli* bacteria are grown in a culture of nutrients which includes glucose and lactose as the main source of carbon-based nutrient.



Which of the following statements accounts for the growth curve showing the reproduction of *E.coli* bacteria?

	P	Q	R
A	Lactose is metabolised	Production of enzymes for lactose metabolism is repressed	Glucose is metabolised
B	Lactose is metabolised	Enzymes for lactose metabolism is produced	Glucose is metabolized
C	Glucose is metabolised	Production of enzymes for lactose metabolism is repressed	Lactose is metabolised
D	Glucose is metabolised	Enzymes for lactose metabolism is produced	Lactose is metabolized

17. IPTG is an analogue of lactose that binds to the *lac* repressor in the same fashion as allolactose. However, it cannot be metabolised by β -galactosidase. *E. coli* cells, which were grown in the absence of lactose and glucose, were initially supplemented with IPTG. After 10 minutes, glucose was added to the cells. 10 minutes later, cAMP was added to the cells. Which of the following graphs represents the amount of *lac* operon mRNA during the time course of this experiment?



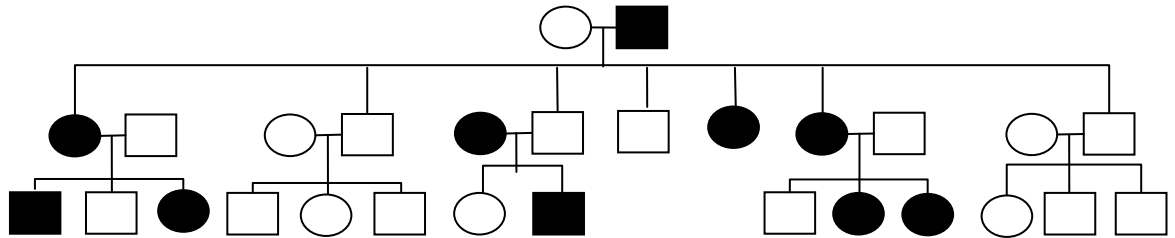
18. Which of the following statements about cancer is / are true?

I	Cancer is a result of increased cell division which promotes the mutation of a proto-oncogene.
II	Individuals who inherit one inactive copy of tumour suppressor gene are more likely to develop cancer than individuals with two non-mutant copies.
III	Mutagenic activation of a single oncogene is sufficient to convert a normal cell into a cancer cell.

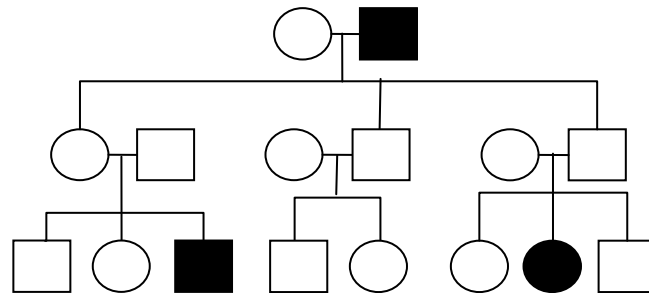
- A II only
 B I and III only
 C II and III only
 D All of the above

19. In 2003, scientists discovered two new genetically transmitted diseases, **P** and **Q**, commonly found among a community living in the South of France. The pedigree diagrams below show the inheritance of these diseases. A general key to these diagrams is also given.

Disease P



Disease Q



Key:

○
unaffected female

●
affected female

□
unaffected male

■
affected male

The alleles responsible for causing disease **P** and **Q** respectively are as follows:

- A** autosomal dominant, sex-linked recessive
- B** autosomal recessive, sex-linked dominant
- C** sex-linked dominant, autosomal recessive
- D** sex-linked recessive, autosomal dominant

20. In mice, the gene for 'dappled' coat (D) and its recessive allele, the gene for 'plain' coat (d), are located on the X chromosome. The gene for 'straight' whiskers (W) and its recessive allele, the gene for 'bent' whiskers (w), are autosomal. A male mouse with plain coat and bent whiskers was mated on several occasions to the same female and the large number of offspring consisted of the following phenotypes in equal proportion:

dappled male with straight whiskers	plain male with straight whiskers
dappled female with straight whiskers	plain female with straight whiskers
dappled male with bent whiskers	plain male with bent whiskers
dappled female with bent whiskers	plain female with bent whiskers

If X^D represents an X chromosome carrying a gene for 'dappled' coat and X^d represents an X chromosome carrying a gene for 'plain' coat, what is the genotype of the female parent?

- A $X^D X^D WW$
- B $X^D X^D Ww$
- C $X^D X^d WW$
- D $X^D X^d Ww$

21. The coat color in labrador retrievers is controlled by two genes, i.e. B/b and E/e. The allele B codes for black coat colour and is completely dominant over allele b which codes for chocolate coat colour. The "expression gene" when dominant ("E") allows the coat to be these colours, but the recessive form (e) results in the darker colours not being expressed resulting in a yellow labrador retriever.

Which of the following statements about the two genes and their effects in the coat colour of Labrador retrievers is true?

- I A yellow labrador retriever must either have the genotype Bbee or bbee
- II A chocolate labrador retriever may have the genotype bbEe.
- III The genotypes of the parents of a black labrador retriever might be BBEE and bbee.
- IV The allele B is completely epistatic to the allele b and the allele E is completely epistatic to allele e.

- A I only
- B II and III only
- C I, II and III only
- D I, II, III and IV

22. The table summarises a breeding experiment with *Drosophila melanogaster*.

parents	BBNN	X	bbnn	
F1	BbNn			
Test cross	BbNn	X	bbnn	
Test cross offspring	BbNn	bbNn	Bbnn	bbnn
No. of test cross offspring	586	111	106	465

Key

B	wild type body colour
b	black body colour
N	normal wings
n	vestigial wings

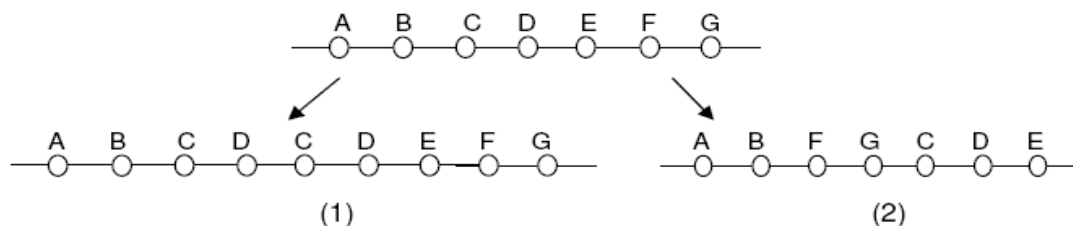
What do these results suggest?

- A** Independent assortment has not occurred.
- B** The alleles N and n are linked.
- C** The alleles for normal and vestigial wings are codominant.
- D** The gene loci for body colour and wing shape are not linked.

23. What correctly describes the genetic basis of continuous variation of the phenotype?

	<i>number of genes involved</i>	<i>effect of gene(s)</i>	
A	many	each gene has a small, additive effect	Quantitative
B	few	each gene has a small, additive effect	Qualitative
C	many	each gene has a large effect	Quantitative
D	few	each gene has a large effect	Quantitative

24. The diagrams below show two types of chromosomal mutation producing changes from the normal gene sequence.



Which of the following shows the two forms of chromosome mutation shown above?

	(1)	(2)
A	Deletion	Duplication
B	Duplication	Translocation
C	Duplication	Inversion
D	Inversion	Duplication

25. The concentration of carbon dioxide in a sample of air was found to be 280 ppm (parts per million). A controlled experiment was designed to measure the concentration of carbon dioxide in the air after it had flowed over the leaves of a green plant. Measurements were taken at a range of light intensities. (RJC 2007)

The following results were obtained:

Light intensity (% of full sunlight)	Concentration of carbon dioxide in air after flowing over leaves (ppm)
75	253
50	252
25	254
10	280

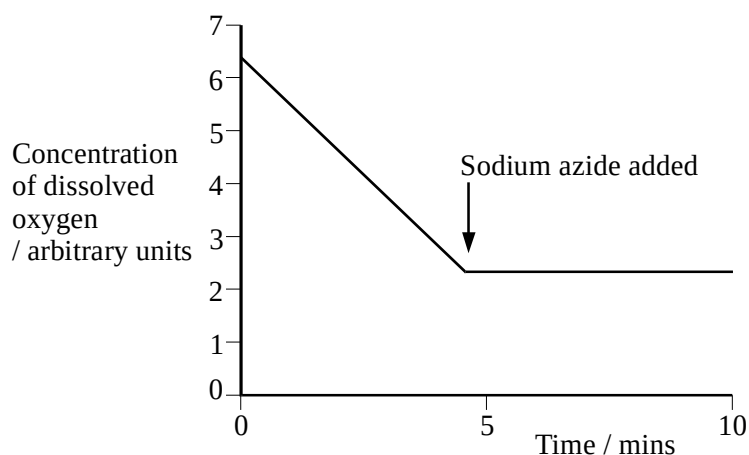
Which one of the following statements is not consistent with these results?

- A** At the lower light intensities tested, the rate of photosynthesis is limited by light intensity.
- B** At the higher light intensities tested, the rate of photosynthesis is affected by factors other than light.
- C** In the dark, the concentration of carbon dioxide in the air after it had flowed over leaves would be at least 280 ppm.
- D** At a light intensity of 10% of full sunlight, photosynthesis does not occur.

26. Active mitochondria can be isolated from liver cells. If these mitochondria are then incubated in a buffer solution containing a substrate, such as succinate, dissolved oxygen will be used by the mitochondria. The concentration of dissolved oxygen in the buffer solution can be measured using an electrode.

An experiment was carried out in which a suspension of active mitochondria was incubated in a buffer solution containing succinate, an intermediate of the Krebs cycle. The concentration of dissolved oxygen was measured every minute for five minutes. A solution containing sodium azide was then added to this preparation and the concentration of dissolved oxygen was measured for a further five minutes.

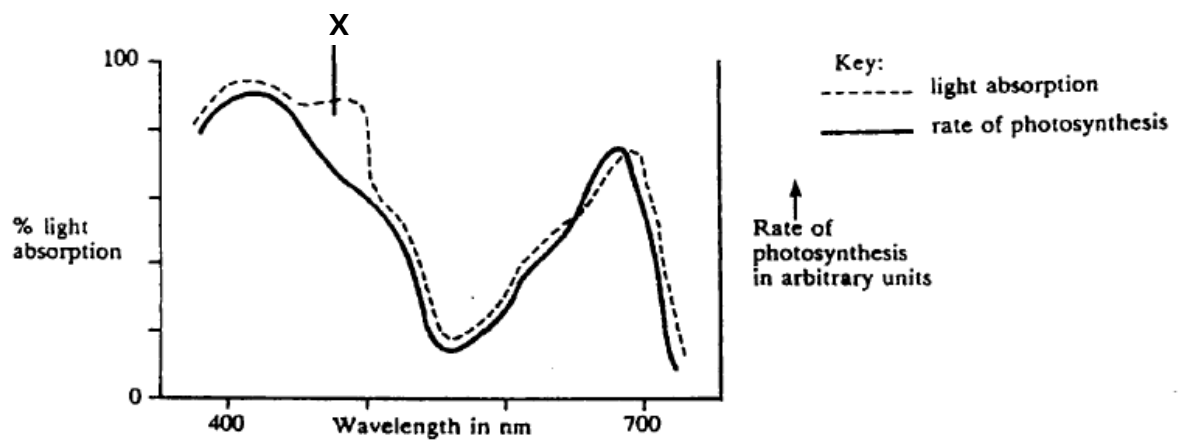
The results are shown in the graph below.



Which of the following statements does not explain the results shown in the graph?

- A The decrease in the concentration of dissolved oxygen in the first 5 minutes is due to oxygen combining with hydrogen ions and electrons to form water.
- B Sodium azide inhibits cytochrome oxidase from binding to water.
- C When sodium azide is added, the electron flow down the electron transport chain is inhibited.
- D Succinate cannot be oxidised when sodium azide is added.

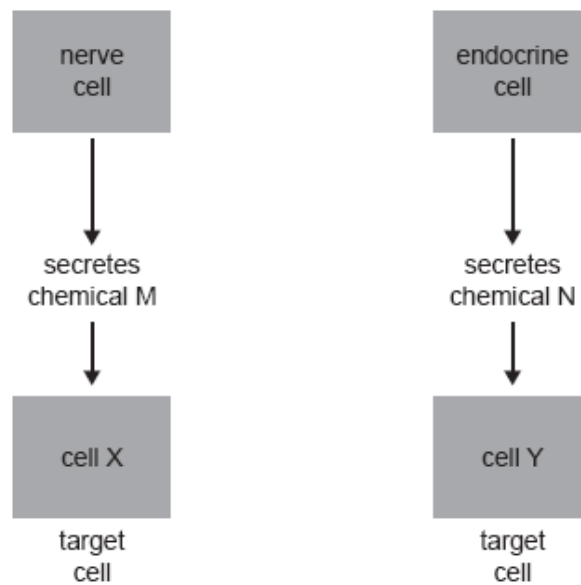
27. The graph below shows the effect of different wavelengths of light on the rate of photosynthesis and on the amount of light absorbed by the pigments in a green seaweed.



The difference between the two curves at X is due to

- A inefficient trapping of light energy by the chlorophyll
 - B no ATP production at that wavelength
 - C oxygen given off during photosynthesis interferes with the absorption of light
 - D carotenes absorbing light that is not used in photosynthesis
28. The *ras* gene codes for the Ras protein which is a G protein. Ras protein activates several pathways that control proliferation and cell migration. When a mutation occurs on the *ras* gene, uncontrolled cell division occurs. This is because
- A the mutation changes the conformation of the Ras protein, preventing it from being activated.
 - B the growth factor is not bound to the tyrosine kinase receptor, hence Ras protein cannot be activated.
 - C the mutation prevents GTP hydrolysis on the Ras protein causing it to be hyperactive.
 - D phosphatases were inhibited hence the protein kinases involved in the phosphorylation cascade could not be inactivated.

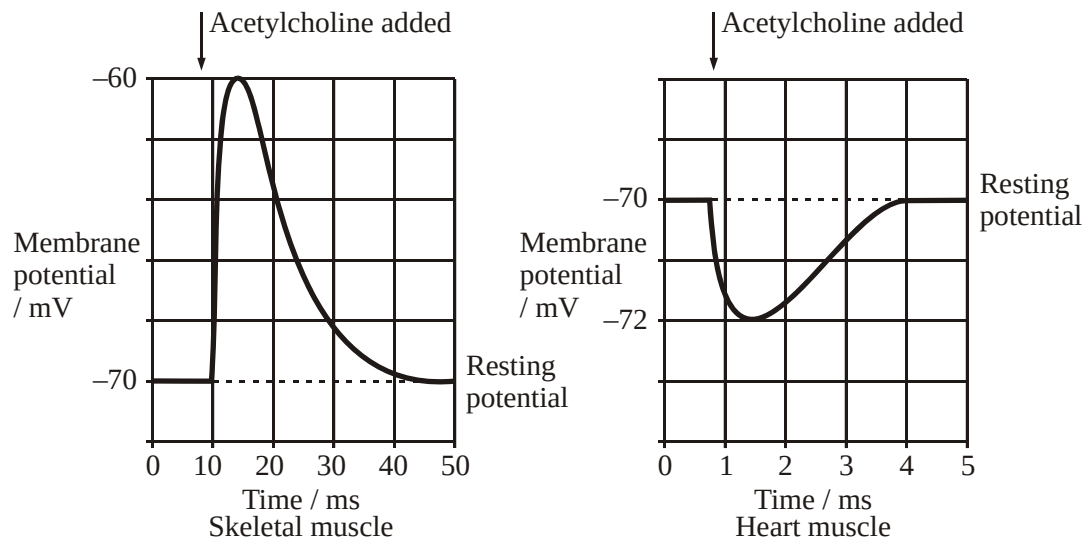
29. The diagram shows how a nerve cell and an endocrine cell interact with cell X and cell Y respectively.



Which of the following is true of both interactions?

- A** Diffusion is involved in the movement of chemicals M and N to their respective target cells.
- B** Chemical M is secreted faster than chemical N.
- C** When both cells X and Y are stimulated, phosphorylation cascades must be initiated.
- D** Both chemicals M and N must bind to cell-surface receptors on cell X and Y respectively.

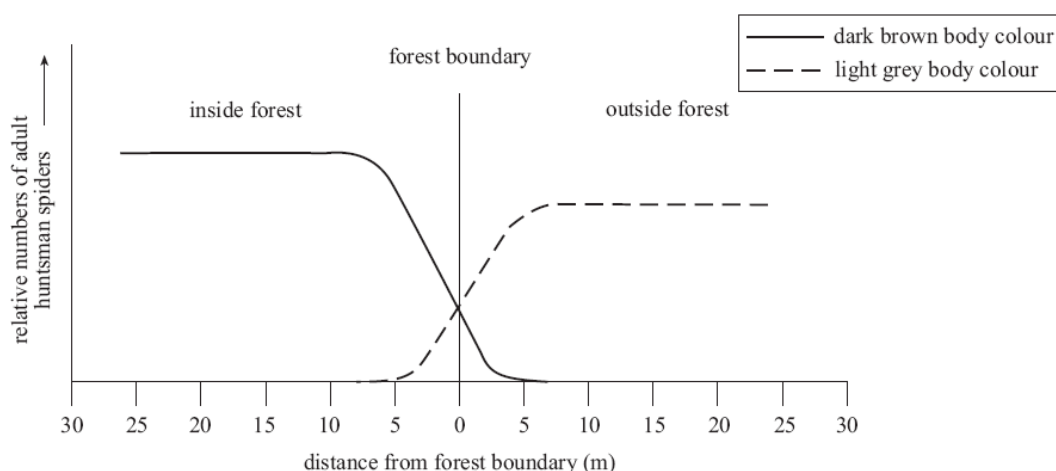
30. The graphs below show the effect of adding acetylcholine to skeletal muscle and heart muscle.



Which of the following statements are true of the results above?

- I Depolarisation occurred when acetylcholine was added to skeletal muscle but hyperpolarisation occurred when acetylcholine was added to heart muscle.
 - II Sodium channels opened when acetylcholine was added to skeletal muscle but potassium channels opened when acetylcholine was added to heart muscle.
 - III Acetylcholine causes contraction of the skeletal muscle and causes a decrease in heart muscle contraction.
 - IV Binding of acetylcholine leads to formation of IPSP in skeletal muscle and EPSP in heart muscle.
- A** I and II only
- B** II and IV only
- C** I, II and III only
- D** I, II, III and IV

31. The graph below shows the distribution of huntsman spiders at a forest boundary.



One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living in the leaf litter inside the forest and others were found living in the grass just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

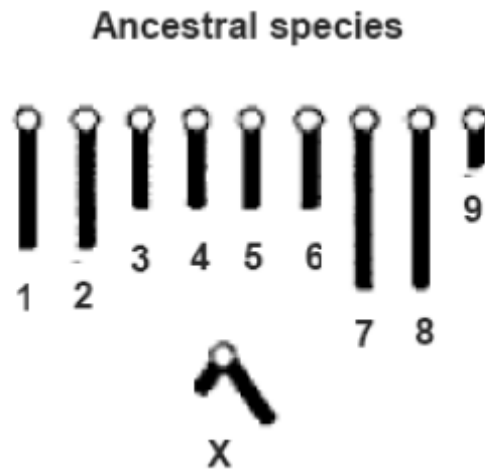
The best explanation for this distribution is that

- A the two populations of spiders were once different species.
- B the two populations of spiders were unable to interbreed and individuals were adapting to suit their habitats.
- C the differences in the two habitats had changed the physical appearance of individual spiders.
- D a particular body colour provided a selective advantage to spiders in a particular habitat.

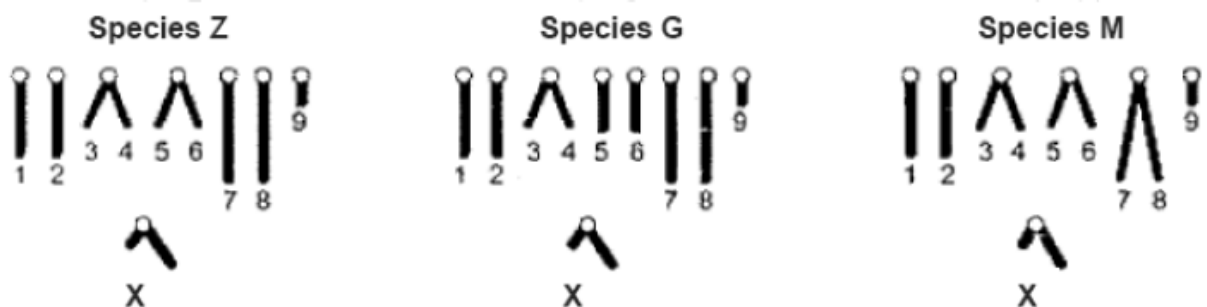
32. Which of the following is an example of mechanical isolation?

- A Two species of frogs have different mating calls.
- B The flower structure of one species prevents the transfer of pollen from another species.
- C Two species of birds live in the same habitat; one mates in spring and the other in summer.
- D One species of lizards inhabits the trees, and another species inhabits the ground cover.

33. In some Australian insects, new species have arisen through changes that occurred to chromosomes in an ancestral species. Such changes may involve the joining together of chromosomes, the loss of whole or parts of chromosomes, and rearrangement of the genetic material within chromosomes. One ancestral species has the following haploid set of chromosomes.



As the changes in chromosomes accumulate, a number of different species can result from a single ancestral species. Three species that have evolved from the ancestral species shown above have the haploid sets of chromosomes shown below.



The most likely order of evolution of these species is

- A ancestral species, species Z, species G, species M.
- B ancestral species, species G, species M, species Z.
- C ancestral species, species M, species G, species Z.
- D ancestral species, species G, species Z, species M.

34. The table shows equivalent amino acid sequences of part of a protein from four species of animal.

animal	amino acid sequence							
1	trp	met	val	glu	cys	asp	arg	leu
2	trp	val	met	glu	cys	asp	asp	ala
3	trp	val	val	glu	cys	asp	arg	leu
4	phe	trp	val	gly	cys	arg	asp	leu

Using the technique of molecular homology, which pair of animals share the most recent common ancestor and which pair are least closely related?

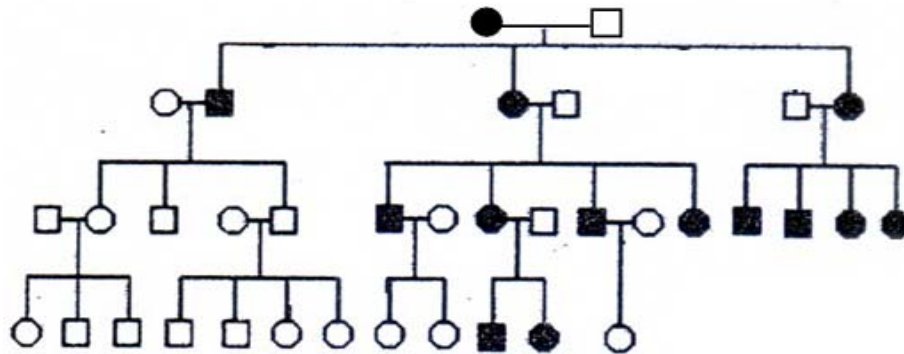
	Most closely related pair	Least closely related pair
A	1 and 2	1 and 4
B	1 and 3	1 and 4
C	1 and 3	2 and 4
D	2 and 3	3 and 4

35. Which uses of the information from the human genome project are generally considered to be unethical?

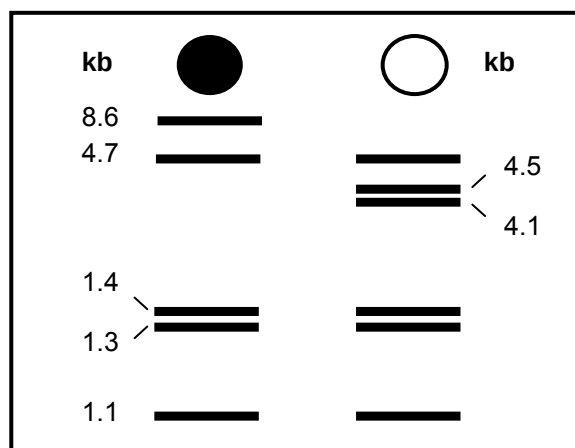
- 1 an insurance company only giving cheap rates to people with genetic predispositions to fewer diseases
- 2 genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships
- 3 cytologists developing tests for only some defective genes
- 4 doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes
- 5 genetic councillors giving specific lifestyle information only to people genetically predisposed to risks
- 6 parents choosing embryos for implantation only after ante-natal tests for acceptable genes

- A** 1 and 3
- B** 1 and 6
- C** 2 and 6
- D** 3 and 4

36. The pedigree shows the inheritance of an RFLP marker locus which contains restriction sites for *Hae*II in mitochondrial DNA (mtDNA). mtDNA exists as a double-stranded circular DNA molecule, very much resembling a bacterial plasmid. The filled symbols indicate individuals who have lost a *Hae*II site to a point mutation.



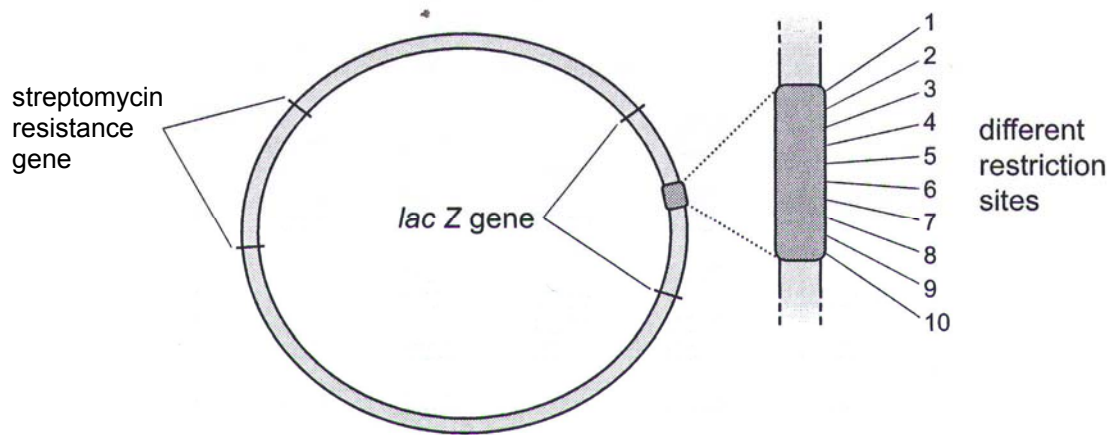
mtDNA from an affected individual and an unaffected individual is digested with *Hae*II to produce restriction fragments that are separated using gel electrophoresis. The results are shown as follows:



Which of the following statements is not correct?

- A There were six *Hae*II restriction sites in the mtDNA before the mutation.
- B Affected females always transmit the trait to all their children.
- C Mitochondrial genes in males are not transmitted to any of their children.
- D The mutation occurred at the restriction sites between fragments of 4.1 kb and 4.7 kb.

37. The diagram shows the plasmid pUC18. Bacteria containing this plasmid produce blue colonies when grown in the presence of *X-gal*. Bacteria containing a genetically engineered recombinant pUC18 plasmid produce white colonies.



Some features of this plasmid are:

- 1 It is small and replicates to form about 500 copies per host cell.
- 2 It contains restriction sites for 10 different restriction enzymes.
- 3 It contains a gene giving resistance to the antibiotic streptomycin.
- 4 It contains the *lac Z* gene which allows the metabolism of *X-gal* to produce a blue colour.

A gene of interest was inserted into one of the restriction sites to form a recombinant plasmid. Bacteria were transformed with this recombinant plasmid and identified using a selective agar medium.

Which selective growth medium would identify the bacteria containing the recombinant plasmids?

- A a medium containing agar
- B a medium containing streptomycin
- C a medium containing streptomycin and *X-gal*
- D a medium containing *X-gal*

38. Which of the following statements regarding the properties of haematopoietic stem cells that makes them useful in gene therapy is not true?
- A** They are pluripotent and have the ability to self-renew, which means that the need to provide repeated administrations of gene therapy can be reduced or possibly even eliminated.
 - B** They can easily be removed from the blood, bone marrow or umbilical cord.
 - C** They can be more readily coaxed to differentiate in a laboratory setting before injection into the patient.
 - D** They are not as difficult to identify and isolate as embryonic stem cells.
39. Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to give infertile triploid salmon. Reproductive organ tissue from diploid trout was transplanted into newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish could be seen as genetically modified?

- A** salmon providing eggs for treatment
 - B** trout providing reproductive organ tissue
 - C** young triploid salmon
 - D** young trout
40. Which genetic modification would enable higher yield in crops grown on fertile soil in a tropical region?
- A** drought tolerance
 - B** increased carbon fixation
 - C** increased vitamin A content
 - D** salt tolerance